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United States Patent Application Publication

20250281995

Kind Code

A1

Publication Date

September 11, 2025

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Flash-Butt Welding Vehicle, Systems, and Methods of Using the Same

Abstract

A single flash-butt welding vehicle performs most if not all operations needed for flash-butt weld repair of a railroad track. Specifically, the flash-butt welding vehicle utilizes primary and secondary integral cranes for moving and manipulating rail and for welding the same. The vehicle further utilizes one or more rail storage areas thereon for loading, transporting, and unloading multiple sizes and lengths of rail. Systems and methods of using the same are further provided.

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Appl. No.: 19/074867

Filed: March 10, 2025

Related U.S. Application Data

us-provisional-application US 63562817 20240308

Publication Classification

Int. Cl.: B23K11/04 (20060101); B23K101/26 (20060101)

U.S. Cl.:

CPC B23K11/046 (20130101); B23K2101/26 (20180801)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present invention claims priority to U.S. Provisional Pat. App. No. 63/562,817, titled “Flash-Butt Welding Vehicle, Systems, and Methods of Using the Same,” filed Mar. 8, 2024, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present invention relates to a flash-butt welding vehicle that performs most if not all operations needed for flash-butt weld repair of a railroad track. Specifically, the flash-butt welding vehicle utilizes primary and secondary integral cranes for moving and manipulating rail and for welding the same. The vehicle further utilizes one or more rail storage areas thereon for loading, transporting, and unloading multiple sizes and lengths of rail. Systems and methods of using the same are further provided.

BACKGROUND

[0003] Flash-butt welding is a type of resistance welding that welds two pieces of metal together and is ideal for use in repairing railroad tracks. Typically, the pieces of metal, i.e., the rail sections to be welded, are set apart at a predetermined distance based on material thickness, material composition, and desired properties of the finished weld. Current is applied to the metal, and the gap between the two pieces creates resistance and produces the arc required to melt the metal. Once the pieces of metal reach the proper temperature, they are pressed together, effectively forge-welding them together.

[0004] As noted, railroads use flash-butt welding to join sections of mainline rail together to create Long Welded Rail (LWR) in a factory setting or continuous welded rail (CWR) in track, which is much smoother than mechanically joined rail because there are no gaps between the sections of rail. Flash-butt welding is also beneficial because it allows dissimilar metals, including non-ferrous metals, to be joined. This allows switches and crossings, which are generally composed of high manganese steel, to be effectively welded to carbon steel rail with the use of a stainless steel insert, while keeping the desired mechanical properties of both the rails and the crossings intact.

[0005] In flash-butt welding, a damaged rail section is often cut and removed from a railroad track forming a gap on the track. An undamaged rail section of roughly the same length, known as a plug, is then placed precisely within the gap, and the terminal ends of the undamaged rail section are welded to the ends of the existing track that terminate at the gap.

[0006] To repair railroad tracks, it is common to send a crew of several workers and several vehicles to a site needing repair. For example, one truck may carry the rail for use in the repair, another truck may carry the cutting and/or welding equipment, and another may carry a crane for unloading plugs and placing them on site for rail repair. This may further require a crew of about eight or more workers. In another example, two trucks may be utilized, one carrying the rail and a crane for moving the rail and another for carrying the cutting and/or welding equipment, and a smaller crew of about 5 or 6 workers that can fit on the two trucks.

[0007] The use of multiple vehicles for loading, transporting, unloading, and welding rail to repair railroad track is inefficient, not cost effective, and environmentally damaging. Specifically, the use of multiple vehicles requires the use of a large number of crew, some just for piloting the vehicles. In addition, the vehicles require fuel and maintenance, which can be costly, and the workers require payment. The use of multiple vehicles further generates pollution and fosters environmental degradation.

[0008] A need, therefore, exists for an improved vehicle for repairing damaged railroad tracks. Specifically, a need exists for an improved vehicle having the capacity for loading, transporting, and unloading rail sections. Moreover, a need exists for an improved vehicle for further having cutting and/or welding equipment thereon for repairing the railroad tracks.

[0009] Further, a need exists for an improved vehicle for repairing damaged railroad tracks that uses a single vehicle for all operations of flash-butt welding of rail sections when repairing railroad

tracks. Specifically, a need exists for an improved vehicle for repairing damaged railroad tracks that may maximize efficiency and cost effectiveness. Moreover, a need exists for an improved vehicle for repairing damaged railroad tracks that is cleaner for the environment than prior systems.

SUMMARY OF THE INVENTION

[0010] The present invention relates to a flash-butt welding vehicle that performs most if not all operations needed for flash-butt weld repair of a railroad track. Specifically, the flash-butt welding vehicle utilizes primary and secondary integral cranes for moving and manipulating rail and for welding the same. The vehicle further utilizes one or more rail storage areas thereon for loading, transporting, and unloading multiple sizes and lengths of rail. Systems and methods of using the same are further provided.

[0011] To this end, in an embodiment of the present invention, a vehicle for use in flash-butt welding rail sections on a railroad track is provided. The vehicle comprises a bed having a first crane mounted thereon, wherein the first crane comprises welding equipment utilized for flash-butt welding, a second crane for deploying rail sections from the vehicle, and at least one storage area for holding, storing, and transporting rail sections.

[0012] It is, therefore, an advantage and objective of the present invention to provide an improved vehicle for repairing damaged railroad tracks.

[0013] Specifically, it is an advantage and objective of the present invention to provide an improved vehicle having the capacity for loading, transporting, and unloading rail sections.

[0014] Moreover, it is an advantage and objective of the present invention to provide an improved vehicle for further having cutting and/or welding equipment thereon for repairing the railroad tracks.

[0015] Further, it is an advantage and objective of the present invention to provide for the repair of damaged railroad tracks that uses a single vehicle for all operations of flash-butt welding of rail sections when repairing railroad tracks.

[0016] Specifically, it is an advantage and objective of the present invention to provide an improved vehicle for repairing damaged railroad tracks that may maximize efficiency and cost effectiveness.

[0017] Moreover, it is an advantage and objective of the present invention to provide an improved vehicle for repairing damaged railroad tracks that is cleaner for the environment than prior systems.

[0018] Additional features and advantages of the present invention are described in, and will be apparent from, the detailed description of the presently preferred embodiments and from the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] The drawing figures depict one or more implementations in accord with the present concepts, by way of example only, not by way of limitations. In the figures, like reference numerals refer to the same or similar elements.

[0020] FIGS. 1A and 1B illustrate a opposite perspective views of a flash-butt welding vehicle in an embodiment of the present invention.

[0021] FIG. 2 illustrates a top view of a flash-butt welding vehicle in an embodiment of the present invention.

DETAILED DESCRIPTION OF THE PRESENTLY PREFERRED EMBODIMENTS

[0022] The present invention relates to a flash-butt welding vehicle that performs most if not all operations needed for flash-butt weld repair of a railroad track. Specifically, the flash-butt welding vehicle utilizes primary and secondary integral cranes for moving and manipulating rail and for welding the same. The vehicle further utilizes one or more rail storage areas thereon for loading,

transporting, and unloading multiple sizes and lengths of rail. Systems and methods of using the same are further provided.

[0023] FIGS. **1A** and **1B** illustrate opposite views of a flash-butt welding vehicle **10** in an embodiment of the present invention. Specifically, the flash-butt welding vehicle **10** generally comprises a cab **12**, a bed **14**, a plurality of wheels **16** for moving the truck on a roadway, and a plurality of rail wheels **18** for moving the vehicle **10** along a railroad track. As such, the vehicle may operate on both roadways and railroad tracks, known in the industry as a dual mode hi-rail truck.

[0024] Disposed on the bed **14** may be a number of elements that may be useful for repairing railroad tracks. Specifically, the bed **14** may hold a first crane **20** having welding equipment **22** thereon, which may be utilized to extend and deploy from the bed **14** and be placed in any adjacent location to weld rail sections together via flash-butt welding.

[0025] Further disposed on the bed **14** may be a second crane **30** which may be utilized for loading, unloading, and placing rail sections from the vehicle **10**, as described in more detail below. The second crane **30** may extend at or near the rear of the bed **14** and may be utilized for loading the rail sections on to the bed **14**, such as at a depot or the like prior to deploying the vehicle **10** to a location for repairing railroad tracks. Moreover, the second crane **30** may remove rail sections from the bed **14** and may further place the rail sections on either side of the vehicle **10**, behind the vehicle **10**, and at many different angles relative to the vehicle **10**. Thus, once a damaged rail section is removed from a railroad track, the second crane **30** may be utilized for unloading a rail section, deploying the rail section for cutting of the same, and for precisely placing the fitted rail section into a gap left when the damaged rail section is removed. FIG. **2** illustrates the second crane **30** deployed to precisely place a rail section behind the vehicle **10**.

[0026] Additionally, the bed **14** may further have one or more storage areas **40**, **42** for storing various lengths of rail for use in flash-butt welding. Preferably, the one or more storage areas **40**, **42** may be disposed beneath the first crane **20** and/or the second crane **30** and may allow various lengths of rail to be stored and transported therein. The storage areas **40**, **42** may be accessible from the rear of the bed **14** so that a rail section may be easily grabbed and pulled therefrom using the second crane **30**. The storage areas **40**, **42** may run the length of the bed **14**, thereby allowing lengthy rail sections **50**, **52** to be loaded, stored, transported, unloaded, and used during a rail repair process. For example, rail sections having lengths of up to about 19' 6" may be loaded, stored, transported, unloaded, and used using the vehicle **10** of the present invention. Of course, any length of rail may be placed thereon and deployed therefrom and the present invention should not be limited as described herein.

[0027] The vehicle **10** may have a plurality of other elements that may be useful for repairing railroad tracks via flash-butt welding. For example, controls and machinery for manipulating the first crane **20** and/or second crane **30** may be located thereon. Moreover, various toolboxes and storage boxes may be mounted thereon, as well as any number of tools may be stored therein as may be necessary to repair a railroad track. In addition, a cutting saw or other cutting mechanism may be present on the bed **14**, such as a saw for cutting rail sections to precise lengths. For example, the saw may be removable for placing on damaged rail to cut therethrough and/or for cutting rail sections to precise lengths for flash-butt welding of the same.

[0028] Because the vehicle **10** is a dual mode hi-rail vehicle, it may be transported to a damaged rail either through a roadway system, on the railroad tracks, or both, allowing the vehicle to have access to any damaged railroad track section necessary. Moreover, the vehicle **10** may further be placed on the tracks via the rail wheels **18** for deployment of the welding equipment **22** on the first crane **20** and/or for deployment of the rail sections using the second crane **30** behind the vehicle **10**. Alternatively, the vehicle may be driven next or adjacent the railroad tracks via the road wheels **16** for deployment of the welding equipment **22** and/or for deployment of the rail sections beside the vehicle **10**.

[0029] It should be noted that various changes and modifications to the presently preferred embodiments described herein will be apparent to those skilled in the art. Such changes and modifications may be made without departing from the spirit and scope of the present invention and without diminishing its attendant advantages. Further, references throughout the specification to “the invention” are nonlimiting, and it should be noted that claim limitations presented herein are not meant to describe the invention as a whole. Moreover, the invention illustratively disclosed herein suitably may be practiced in the absence of any element which is not specifically disclosed herein.

Claims

1. A vehicle for use in flash-butt welding rail sections on a railroad track comprising: a bed having a first crane arm mounted thereon, wherein the first crane arm comprises welding equipment utilized for flash-butt welding; a second crane arm for deploying rail sections from the vehicle; and a first storage area for holding, storing, and transporting one or more railroad track rail sections.
 2. The vehicle of claim 1 further comprising: a plurality of wheels for driving the vehicle on a roadway; and a plurality of train wheels for driving the vehicle on a railroad track.
 3. The vehicle of claim 1 wherein the first crane arm is configured to extend and deploy from the vehicle.
 4. The vehicle of claim 1 wherein the second crane arm is configured to extend and deploy from the vehicle.
 5. The vehicle of claim 1 wherein the second crane arm is configured to load, unload, and place rail sections from the vehicle.
 6. The vehicle of claim 1 wherein the second crane arm extends from a location at or near a rear of the bed.
 7. The vehicle of claim 1 wherein the storage area is disposed beneath the first crane arm or the second crane arm.
 8. The vehicle of claim 1 wherein the first storage area has an opening for loading and unloading the railroad track rail sections.
 9. The vehicle of claim 8 wherein the opening is disposed at or near a rear of the bed.
 10. The vehicle of claim 1 further comprising: a second storage area within the bed.
 11. The vehicle of claim 1 wherein the first storage area and the second storage area are disposed at or near the rear of the bed.
 12. The vehicle of claim 1 wherein the first storage area and the second storage area are disposed on opposite sides of the bed.
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